

Amazon Data Engineer - 6 Month Study Plan

Month 1: Programming & DSA (20%)

Topics: Python, Data Structures, Algorithms YouTube:

<https://www.youtube.com/watch?v=rfscVS0vtbw> Docs: <https://docs.python.org/3/tutorial/> GitHub:

<https://github.com/TheAlgorithms/Python> Tip: Practice daily on LeetCode

Month 2: SQL & Data Modeling (25%)

Topics: Advanced SQL, Window Functions, Indexing YouTube:

<https://www.youtube.com/watch?v=HXV3zeQKqGY> Docs: <https://www.postgresql.org/docs/>

GitHub: <https://github.com/borall/sql-anti-patterns> Tip: Solve real datasets

Month 3: Data Engineering Fundamentals (20%)

Topics: ETL, Pipelines, Airflow YouTube: <https://www.youtube.com/watch?v=AHMm1wfGuHE>

Docs: <https://airflow.apache.org/docs/> GitHub: <https://github.com/apache/airflow> Tip: Build projects

Month 4: Big Data (15%)

Topics: Spark, Hadoop YouTube: https://www.youtube.com/watch?v=_C8kWso4ne4 Docs:

<https://spark.apache.org/docs/latest/> GitHub: <https://github.com/apache/spark> Tip: Practice transformations

Month 5: AWS (15%)

Topics: S3, Redshift, Glue YouTube: <https://www.youtube.com/watch?v=ulprqHHWIng> Docs:

<https://docs.aws.amazon.com/> GitHub: <https://github.com/aws-samples> Tip: Do hands-on labs

Month 6: System Design & Interview (5%)

Topics: Data System Design, Mock Interviews YouTube:

https://www.youtube.com/watch?v=UzLMhgg3_Wc Docs:

<https://github.com/donnemartin/system-design-primer> GitHub:

<https://github.com/donnemartin/system-design-primer> Tip: Practice explanations